

Rank-One P_2 -Scale Prime-Tail Resummation and Bounded-Gap Necessity

RH research program

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Abstract

Let $x = p_y$, $L = \log x \geq 512$, and retain the exact first strict prime tail $P_1(y)$. For every integer $1 \leq K \leq \lfloor 3L \rfloor$, we replace all higher tails by the factorial Laplace kernel

$$I_{2r}^{[K]} = \frac{x^{1-2r}}{(2r-1)L} \sum_{j=0}^{K-1} \frac{(-1)^j j!}{((2r-1)L)^j} \quad (r \geq 2).$$

For the seven endpoint channels and the exact RH-383 endpoint map, the resulting rank-one surrogate Gap_K satisfies

$$\pi^2 |\text{Gap}_P - \text{Gap}_K| \leq \frac{x^{-3}}{L} \left(7560\varepsilon_x + \frac{1638}{x^2} + 1176 \frac{K!}{(3L)^K} \right),$$

where $\varepsilon_x = 0.027L^{1.801} \exp[-0.1853L^{3/5}(\log L)^{-1/5}]$. The factorial ratio is at most $1/(3L)$ throughout the full integer window, so the maximum over all such K is $o(P_2(y))$.

This rank-one retention is sharp within the frozen $P/J/I$ kernel hierarchy. Maynard's theorem on infinitely many bounded consecutive prime gaps implies

$$\limsup_y p_y^2 |P_1(y) - I_2(p_y)| \geq \frac{1}{2}$$

and, with $X_\infty > 0$ from the exact endpoint,

$$\limsup_y p_y^2 \pi^2 |\text{Gap}_P - \text{Gap}_I| \geq X_\infty, \quad \limsup_y p_y^2 \pi^2 |\text{Gap}_P - \text{Gap}_J| \geq X_\infty.$$

Thus a fully smooth J/I surrogate cannot attain the P_2 scale, whereas preserving the exact P_1 coordinate does. The result is not a claim that the factorial series converges and not a universal impossibility theorem for every conceivable surrogate. A 56-row exact artifact reproduces the algebraic interfaces but is not an analytic proof.

Keywords: bounded consecutive prime gaps; strict prime tails; factorial asymptotics; rank-one retention; square-clock endpoint; P_2 scale.

1 Setting, inherited endpoint, and main results

Let $3 = p_1 < p_2 < \dots$ be the odd primes and put

$$x = p_y, \quad L = \log x, \quad V(L) = L^{3/5}(\log L)^{-1/5}, \quad \varepsilon_x = \frac{27}{1000} L^{1801/1000} e^{-(1853/10000)V(L)}.$$

Johnston and Yang's Theorem 1.4, equation (1.8), printed page 2, gives

$$|\vartheta(t) - t| \leq t\varepsilon_t \quad (t \geq 23), \tag{1}$$

and RH-386 proves that the displayed envelope decreases once $\log t \geq 512$ [1, 6]. Hence $\varepsilon_t \leq \varepsilon_x$ for $t \geq x$ in our domain.

For $r \geq 1$ define the strict prime and integral tails

$$P_r(y) = \sum_{p>x} (p^2 - 1)^{-r}, \quad J_r(x) = \int_x^\infty \frac{dt}{(t^2 - 1)^r \log t}, \quad I_{2r}(x) = \int_x^\infty \frac{t^{-2r}}{\log t} dt. \quad (2)$$

For $c = 1, \dots, 7$ let

$$\Phi_c^P = \sum_{r \geq 1} \frac{c^r P_r}{r}, \quad \Phi_c^J = \sum_{r \geq 1} \frac{c^r J_r}{r}, \quad \Phi_c^I = \sum_{r \geq 1} \frac{c^r I_{2r}}{r}. \quad (3)$$

The nonnegative Tonelli identities identify Φ_c^J and Φ_c^I with the logarithmic integrals in RH-387 [7].

For $r \geq 2$, set

$$K_r = \frac{x^{1-2r}}{(2r-1)L}, \quad a_r = \frac{1}{(2r-1)L}, \quad S_K(a) = \sum_{j=0}^{K-1} (-1)^j j! a^j, \quad (4)$$

and

$$I_{2r}^{[K]} = K_r S_K(a_r), \quad \Psi_c^{[K]} = c P_1(y) + \sum_{r \geq 2} \frac{c^r I_{2r}^{[K]}}{r}. \quad (5)$$

The first coordinate is therefore retained exactly; every rank $r \geq 2$ is replaced.

We recall the exact endpoint map. For $2 \leq m \leq 8$, put

$$u_m = \prod_{p \text{ odd}} \frac{1 - m/p^2}{1 - 1/p^2},$$

and index

$$(\alpha_m) = (-2, 2, -2, 2, -2, 2, -2), \quad (\beta_m) = (1, -2, 2, -2, 2, -2, 2).$$

With

$$C(Z) = 1 + \sum_{m=2}^8 \alpha_m Z_m, \quad W(Z) = \sum_{m=2}^8 \beta_m Z_m, \quad Z_m(z) = u_m e^{z_{m-1}},$$

define

$$F(z) = 2\{C(u) - C(Z(z))\} - 4W(Z(z))(1 - e^{-z_1}). \quad (6)$$

The exact normal form of RH-383 [4] gives, for $q_y = 4 \prod_{i \leq y} p_i^2$,

$$\text{Gap}_P := B_\infty - G(q_y) = \frac{F(\Phi^P)}{\pi^2}, \quad \text{Gap}_J := \frac{F(\Phi^J)}{\pi^2}, \quad \text{Gap}_I := \frac{F(\Phi^I)}{\pi^2}, \quad \text{Gap}_K := \frac{F(\Psi^{[K]})}{\pi^2}. \quad (7)$$

Here each undecorated coordinate symbol denotes its seven-vector. Only Gap_P is the actual square-clock gap; the other three are declared surrogates.

Theorem 1.1 (Uniform rank-one P_2 -scale resummation). *Let $x = p_y$, $L \geq 512$, and let K be any integer satisfying $1 \leq K \leq \lfloor 3L \rfloor$. The vectors Φ^P, Φ^J, Φ^I , and $\Psi^{[K]}$ belong to $[0, 1/2]^7$, and*

$$\max_{1 \leq c \leq 7} |\Phi_c^P - \Psi_c^{[K]}| \leq \frac{x^{-3}}{L} \left(60\varepsilon_x + \frac{13}{x^2} + \frac{28}{3} \frac{K!}{(3L)^K} \right). \quad (8)$$

Consequently,

$$\pi^2 |\text{Gap}_P - \text{Gap}_K| \leq \frac{x^{-3}}{L} \left(7560\varepsilon_x + \frac{1638}{x^2} + 1176 \frac{K!}{(3L)^K} \right). \quad (9)$$

Moreover,

$$\lim_{y \rightarrow \infty} \max_{1 \leq K \leq \lfloor 3 \log p_y \rfloor} \frac{|\text{Gap}_P - \text{Gap}_K|}{P_2(y)} = 0. \quad (10)$$

Define the positive endpoint constant

$$X_\infty = 2u_4 - 4u_5 + 6u_6 - 8u_7 + 10u_8 > 0. \quad (11)$$

Its positivity is the exact run-count consequence proved in RH-381 [3].

Theorem 1.2 (Bounded-gap necessity in the frozen hierarchy). *Within the $P/J/I$ hierarchy defined above,*

$$\limsup_{y \rightarrow \infty} p_y^2 |P_1(y) - I_2(p_y)| \geq \frac{1}{2}, \quad (12)$$

$$\limsup_{y \rightarrow \infty} p_y^2 |P_1(y) - J_1(p_y)| \geq \frac{1}{2}, \quad (13)$$

$$\limsup_{y \rightarrow \infty} p_y^2 \pi^2 |\text{Gap}_P - \text{Gap}_I| \geq X_\infty, \quad (14)$$

$$\limsup_{y \rightarrow \infty} p_y^2 \pi^2 |\text{Gap}_P - \text{Gap}_J| \geq X_\infty. \quad (15)$$

In particular,

$$\limsup_y \frac{|\text{Gap}_P - \text{Gap}_I|}{P_2(y)} = \limsup_y \frac{|\text{Gap}_P - \text{Gap}_J|}{P_2(y)} = +\infty. \quad (16)$$

Thus the exact P_1 term is necessary for the specific smooth-surrogate P_2 contract in Theorem 1.1. No statement is made about every conceivable surrogate.

2 Higher-rank source and power transfers

Write $E(t) = \vartheta(t) - t$ and $h_r(t) = (t^2 - 1)^{-r} / \log t$. The strict endpoint $p > x$ gives

$$P_r - J_r = -E(x)h_r(x) - \int_x^\infty E(t)h'_r(t) dt, \quad |P_r - J_r| \leq \varepsilon_x \{2xh_r(x) + J_r\}. \quad (17)$$

The boundary $-E(x)h_r(x)$ and the second copy of $xh_r(x)$ produced by integration by parts are distinct. We apply (17) only for $r \geq 2$.

Let

$$R(z) = -\log(1 - z) - z = \sum_{r \geq 2} \frac{z^r}{r}. \quad (18)$$

For $0 \leq z < 1$,

$$0 \leq R(z) \leq \frac{z^2}{2(1 - z)}, \quad R'(z) = \frac{z}{1 - z}. \quad (19)$$

The first estimate retains the divisor 2; the second will control the change from $(t^2 - 1)^{-1}$ to t^{-2} .

Proposition 2.1 (Higher-rank coordinate ledger). *For every $c \in \{1, \dots, 7\}$ and $L \geq 512$,*

$$\left| \sum_{r \geq 2} \frac{c^r (P_r - J_r)}{r} \right| < \frac{60\varepsilon_x}{x^3 L}, \quad (20)$$

$$0 \leq \sum_{r \geq 2} \frac{c^r (J_r - I_{2r})}{r} < \frac{13}{x^5 L}. \quad (21)$$

Proof. Since $L \geq 512$,

$$x = e^L > 2^L \geq 2^{512} > 256. \quad (22)$$

All summands are absolutely summable, so Tonelli may be applied to their nonnegative majorants. Summing the boundary term in (17) and using (19) gives

$$2x \sum_{r \geq 2} \frac{c^r h_r(x)}{r} \leq \frac{x^{-3}}{L} \frac{c^2}{(1-x^{-2})^2 \{1 - c/(x^2 - 1)\}}.$$

The J_r contribution obeys

$$\sum_{r \geq 2} \frac{c^r J_r}{r} = \int_x^\infty \frac{R(c/(t^2 - 1))}{\log t} dt \leq \frac{x^{-3}}{L} \frac{c^2}{6(1-x^{-2})^2 \{1 - c/(x^2 - 1)\}}.$$

The reciprocal denominator common to these two terms and to the power comparison below is

$$D_c(x)^{-1} := \frac{1}{(1-x^{-2})^2 \{1 - c/(x^2 - 1)\}} = \frac{1}{(1-x^{-2}) \{1 - (1+c)/x^2\}}. \quad (23)$$

It increases with c and decreases with x , so

$$D_c(x)^{-1} < D_7(256)^{-1} = \frac{536870912}{536797185} < \frac{36}{35} < \frac{5}{4}. \quad (24)$$

The source coefficient is therefore

$$\frac{7}{6} c^2 D_c(x)^{-1} < \frac{6}{5} c^2 \leq \frac{294}{5} < 60,$$

which proves (20) by explicit arithmetic.

For the second comparison, the mean-value theorem and $R'(z) = z/(1-z)$ give

$$0 \leq R\left(\frac{c}{t^2 - 1}\right) - R\left(\frac{c}{t^2}\right) \leq \frac{c^2}{t^2(t^2 - 1)(t^2 - 1 - c)}.$$

Therefore

$$\sum_{r \geq 2} \frac{c^r (J_r - I_{2r})}{r} \leq \frac{x^{-5}}{L} \frac{c^2}{5(1-x^{-2}) \{1 - (1+c)/x^2\}}.$$

By (24), the last coefficient is

$$\frac{c^2}{5} D_c(x)^{-1} < \frac{c^2}{4} \leq \frac{49}{4} < 13.$$

This proves (21) by explicit arithmetic. □

3 Exact factorial remainder, cube, and moving window

The power kernel has a Laplace representation with an exact finite remainder.

Lemma 3.1 (Factorial Laplace identity). *For $a > 0$, let*

$$G(a) = \int_0^\infty \frac{e^{-v}}{1 + av} dv.$$

Then $I_{2r} = K_r G(a_r)$ and, for every integer $K \geq 1$,

$$G(a) - S_K(a) = (-a)^K \int_0^\infty \frac{e^{-v} v^K}{1 + av} dv. \quad (25)$$

Consequently

$$(-1)^K (I_{2r} - I_{2r}^{[K]}) \geq 0, \quad |I_{2r} - I_{2r}^{[K]}| \leq K_r a_r^K K!. \quad (26)$$

Proof. Substitute $t = xe^u$ in I_{2r} and then set $v = (2r - 1)u$. This gives $I_{2r} = K_r G(a_r)$. The finite geometric identity

$$\frac{1}{1 + av} = \sum_{j=0}^{K-1} (-av)^j + \frac{(-av)^K}{1 + av}$$

may be integrated termwise because it is finite. Since $\int_0^\infty e^{-v} v^j dv = j!$, it gives (25); positivity of its integral and $1/(1 + av) \leq 1$ give (26). $\square$

Proposition 3.2 (Factorial coordinate and cube). *If $1 \leq K \leq \lfloor 3L \rfloor$, then*

$$\max_{1 \leq c \leq 7} \left| \sum_{r \geq 2} \frac{c^r (I_{2r} - I_{2r}^{[K]})}{r} \right| \leq \frac{28}{3} \frac{x^{-3}}{L} \frac{K!}{(3L)^K}. \quad (27)$$

Moreover $0 < S_K(a_r) \leq 1$ for every $r \geq 2$, and $\Psi^{[K]}$ belongs to $[0, 1/2]^7$.

Proof. Lemma 3.1, $2r - 1 \geq 3$, and $1/r \leq 1/2$ give

$$\begin{aligned} \sum_{r \geq 2} \frac{c^r}{r} |I_{2r} - I_{2r}^{[K]}| &\leq \frac{x^{-3}}{L} \frac{K!}{(3L)^K} \frac{c^2}{6(1 - c/x^2)} \\ &< \frac{4c}{3} \frac{x^{-3}}{L} \frac{K!}{(3L)^K}. \end{aligned}$$

Maximizing $4c/3$ over $c \leq 7$ proves (27). This sum is over all $r \geq 2$, not only $r = 2$.

The magnitudes $j!a_r^j$ in $S_K(a_r)$ have successive ratio $(j + 1)a_r$. Within the stated window every ratio that occurs is at most $(K - 1)/(3L) < 1$. Pairing the decreasing alternating terms, with the case $K = 1$ immediate, gives $0 < S_K(a_r) \leq 1$.

Finally,

$$P_1(y) \leq \sum_{n > x} \frac{1}{n^2 - 1} = \frac{1}{2} \left(\frac{1}{x} + \frac{1}{x + 1} \right),$$

whereas

$$0 \leq \sum_{r \geq 2} \frac{c^r I_{2r}^{[K]}}{r} \leq \frac{c^2}{6x^3 L (1 - c/x^2)}.$$

At $x > 256$, $L \geq 512$, their sum after multiplication of the first term by $c \leq 7$ is below $1/2$. This proves the cube assertion. $\square$

Lemma 3.3 (Uniform integer window). *For every integer $1 \leq K \leq \lfloor 3L \rfloor$,*

$$b_K := \frac{K!}{(3L)^K} \leq b_1 = \frac{1}{3L}. \quad (28)$$

Proof. For $K + 1 \leq \lfloor 3L \rfloor$,

$$\frac{b_{K+1}}{b_K} = \frac{K + 1}{3L} \leq 1.$$

Induction from b_1 proves the assertion over the complete integer window. Finite checks of small K are only regression fixtures and are not used as a proof of this quantified statement. $\square$

4 Endpoint derivative ledgers and proof of sufficiency

The factors in u_m have summable deficits, and the $p = 3$ factor gives

$$0 < u_m \leq \frac{9-m}{8} \quad (2 \leq m \leq 8). \quad (29)$$

Thus the endpoint arrays satisfy

$$A := \sum_{m=2}^8 |\alpha_m| u_m \leq 7, \quad B := \sum_{m=2}^8 |\beta_m| u_m \leq \frac{49}{8}. \quad (30)$$

Lemma 4.1 (Endpoint gradient and Hessian). *For every $z \in [0, 1/2]^7$,*

$$\|\nabla F(z)\|_1 < 126, \quad (31)$$

$$\sum_{i,j=1}^7 |\partial_{ij} F(z)| < 224. \quad (32)$$

In particular,

$$|F(z) - \nabla F(0) \cdot z| \leq 112 \|z\|_\infty^2. \quad (33)$$

Proof. For the gradient, the derivative of the C term, the derivative falling on W , and the derivative falling on $1 - e^{-z_1}$ have coefficients 2, 4, 4. Since $e^{z_j} \leq e^{1/2} < 2$, their entrywise ledger is

$$\|\nabla F(z)\|_1 \leq e^{1/2}(2A + 4B + 4B) \leq 63e^{1/2} < 126.$$

This is the dual ℓ^1 gradient for an ℓ^∞ input.

For the Hessian, the C term contributes $2A$ before the exponential factor. In the product of W and the loss $1 - e^{-z_1}$, the Hessian of W , the two gradient cross terms, and the Hessian of the loss contribute $4B, 8B, 4B$. Hence

$$\sum_{i,j} |\partial_{ij} F(z)| \leq e^{1/2} \{2A + (4 + 8 + 4)B\} \leq 112e^{1/2} < 224.$$

Taylor's theorem on the segment from 0 to z gives (33), including the factor $1/2$. $\square$

Proof of Theorem 1.1. The cube assertion for Φ^P, Φ^J, Φ^I is the direct integer-tail and integral estimate of RH-387; Proposition 3.2 proves it for $\Psi^{[K]}$. Insert between Φ_c^P and $\Psi_c^{[K]}$ the two coordinates

$$cP_1 + \sum_{r \geq 2} \frac{c^r J_r}{r}, \quad cP_1 + \sum_{r \geq 2} \frac{c^r I_{2r}}{r}.$$

Propositions 2.1 and 3.2 give (8).

The segment from Φ^P to $\Psi^{[K]}$ lies in the convex cube. The ℓ^∞/ℓ^1 mean-value estimate in Lemma 4.1 therefore multiplies the three coordinate constants by 126:

$$126 \cdot 60 = 7560, \quad 126 \cdot 13 = 1638, \quad 126 \cdot \frac{28}{3} = 1176.$$

Together with (7), this proves (9).

RH-384 proves the intrinsic scale

$$P_2(y) \sim \frac{1}{3x^3 L} \quad (34)$$

[5]. Lemma 3.3 makes the right side of (9) uniform in all admissible K . Since $\varepsilon_x \rightarrow 0$, $x^{-2} \rightarrow 0$, and $L^{-1} \rightarrow 0$, division by (34) proves (10). $\square$

5 Bounded consecutive gaps and the rank-one obstruction

Maynard's unconditional Theorem 1.3, printed page 385 (PDF page 3), states

$$\liminf_{n \rightarrow \infty} (p_{n+1} - p_n) \leq 600 \quad (35)$$

[2]. We use only the resulting infinitely many *consecutive* prime pairs $x = p_y$, $q = p_{y+1} = x + h$ with $h \leq 600$. Indeed the gaps are integers: if only finitely many were at most 600, every sufficiently late gap would be at least 601, forcing the liminf to be at least 601.

Lemma 5.1 (Successor jump). *Let $E_y = P_1(y) - I_2(p_y)$. Along the bounded consecutive gaps supplied by (35),*

$$E_y - E_{y+1} = \frac{1}{q^2 - 1} - \int_x^q \frac{dt}{t^2 \log t}, \quad x^2(E_y - E_{y+1}) \rightarrow 1. \quad (36)$$

Consequently $\limsup_y p_y^2 |E_y| \geq 1/2$.

Proof. The exact strict successor identity is $P_1(y) = (q^2 - 1)^{-1} + P_1(y + 1)$. Subtract the two integral tails to get the first formula. Since $h \leq 600$, $q/x \rightarrow 1$ and

$$\frac{x^2}{q^2 - 1} \rightarrow 1, \quad 0 \leq x^2 \int_x^q \frac{dt}{t^2 \log t} \leq \frac{h}{\log x} \rightarrow 0.$$

Thus the scaled jump tends to one. The triangle inequality at the two endpoints, together with $q/x \rightarrow 1$, shows that at least one endpoint of each such pair has its own normalized absolute value at least $1/2 - o(1)$. Infinitely many pairs supply infinitely many distinct endpoints, proving the limsup assertion. An eventual $1/16$ lower witness is a safe finite certificate row, not the sharp constant. $\square$

The same conclusion holds with J_1 because

$$0 \leq J_1(x) - I_2(x) = \int_x^\infty \frac{dt}{t^2(t^2 - 1) \log t} = O\left(\frac{1}{x^3 L}\right) = o(x^{-2}). \quad (37)$$

We now lift the scalar jump through the full endpoint. Set $D_y = \Phi^P(y) - \Phi^I(p_y)$. Exact successors and Tonelli give, in channel c ,

$$(D_y - D_{y+1})_c = -\log\left(1 - \frac{c}{q^2 - 1}\right) - \int_x^q \frac{-\log(1 - c/t^2)}{\log t} dt. \quad (38)$$

Uniformly for $c = 1, \dots, 7$, the first term is $c/q^2 + O(q^{-4})$, while the scaled integral is $O(h/L)$. Hence, along the same bounded gaps,

$$x^2(D_y - D_{y+1}) \rightarrow (1, 2, 3, 4, 5, 6, 7). \quad (39)$$

The endpoint direction is exact. From (6),

$$\nabla F(0) \cdot (1, 2, 3, 4, 5, 6, 7) = \sum_{m=2}^8 u_m \{-2\alpha_m(m-1) - 4\beta_m\} \quad (40)$$

$$= 4u_4 - 8u_5 + 12u_6 - 16u_7 + 20u_8 = 2X_\infty. \quad (41)$$

The individual coefficients for $m = 2, \dots, 8$ are $0, 0, 4, -8, 12, -16, 20$.

Proof of Theorem 1.2. Lemma 5.1 proves (12), and (37) proves (13). The direct integral bounds of RH-387 give $\Phi^I = O(1/(xL))$. Its source comparison gives $\Phi^P = O(1/(xL))$ as well. Lemma 4.1 therefore implies

$$F(\Phi^P) - F(\Phi^I) = \nabla F(0) \cdot (\Phi^P - \Phi^I) + o(x^{-2}),$$

because $x^2\|\Phi^P\|_\infty^2 + x^2\|\Phi^I\|_\infty^2 = O(L^{-2})$.

Equations (39) and (41) show that the scalar sequence $\nabla F(0) \cdot D_y$ has scaled successor jump tending to $2X_\infty$. The same two-endpoint triangle argument as in Lemma 5.1 yields limsup at least X_∞ . The Taylor remainder is $o(x^{-2})$, so (7) proves (14).

RH-387 gives $\pi^2|\text{Gap}_J - \text{Gap}_I| \leq 588/(x^3L) = o(x^{-2})$, which transfers the same lower bound to (15). Finally divide these subsequence lower bounds by (34); since $xL \rightarrow \infty$, the two limsup ratios are infinite. This proves the theorem. $\square$

6 Novelty, limitations, and claim firewall

RH-387 replaced every rank, including P_1 , by a smooth integral. Its source error $\varepsilon_x/(xL)$ is larger than the P_2 scale because $\varepsilon_x x^2 \rightarrow \infty$. The present theorem does not sharpen that source estimate. Instead it keeps P_1 exact and applies the source transfer only to $r \geq 2$, gaining the factor x^{-2} needed for a uniform P_2 -scale result. The exact Laplace remainder simultaneously compiles every $r \geq 2$ and every integer K in a moving window; this is not a finite-partition corollary of RH-386 or RH-387.

The bounded-gap theorem supplies the complementary negative edge: within the declared $P/J/I$ hierarchy, smoothing the rank-one coordinate creates an actual-prime jump of order x^{-2} , much larger than P_2 . This establishes necessity for this hierarchy only. It does not exclude surrogates carrying different exact prime data or other correction terms.

The finite sums S_K are asymptotic truncations. We claim neither that $\sum_j (-1)^j j! a^j$ converges nor that the window may be enlarged beyond $K \leq \lfloor 3L \rfloor$. The paid error gives no P_3 or cubic coefficient precision. Channels are the seven real integers; there is no complex- c theorem. The finite clock is fixed before the prefix limit, $c_{11} = 0$ remains phasewise, and neither the clock nor an order K_N grows with the prefix length. No active- c_{11} cancellation, operator, trace, prime-power trace, zeta-zero identification, or proof of the Riemann hypothesis is supplied. Gates A–E are all false.

7 Executable reproduction and source integrity

The exact artifact has epistemic role `reproduction_not_analytic_proof`. Its 56 rows comprise 12 analytic interfaces, seven channel ledgers, 12 finite factorial regressions, seven endpoint rows, ten bounded-gap/Taylor rows, and eight master ledgers. The canonical certificate has 14,531 bytes and SHA-256

373d870847bb0bf134aa1eba30c5e4d2c3a01dba470af9c75ebacadd81976371.

It independently recomputes the exact fractions, the universal b_K induction, the 60, 13, 28/3 coordinate constants, the 126, 224, 112 endpoint ledgers, and the 7560, 1638, 1176 products. Twenty-four genuine semantic mutations are rejected without calling the fresh certificate or any row builder. Strict JSON parsing, exact Python types, optimized `-O0` replay, and a closed official Draft 2020–12 schema are tested. These finite checks do not prove (1), Maynard’s theorem, Stieltjes integration, Tonelli’s theorem, or the asymptotic limits.

The immutable Git closure contains 77 blobs from the RH-387 release commit

dedd8e8d2c44564e66524a646f9cf5fb9a389c77.

Two canonical remote objects, Johnston–Yang and Maynard, give 79 logical sources. The ordered Git and logical digests are

```
d7f2ee43f56631c8f3442db8fcc6fb423a801b5af7607351623cd449a92c3f73
bffce602d6e3b568eb96662820f08aa457ff5d0de4065f3c9eeac53d8d8dfa39.
```

The canonical remote-lock digests, in source-key order, are

```
d53b93212b7c5b5b6b3f7e890099c48ce8e35f2bff9bdd49f9c330a9b5039786
bd4aad4b7042218e5733bb07db2a513770710628a8ac52d5bcc9881fcb0b5d2e.
```

Neither external PDF nor the Johnston–Yang source tar is vendored or archived. Network verification is disabled by default and requires an explicit opt-in. Offline semantic verification freezes the Maynard article page and DOI. The opt-in retrieval gate checks only the exact requested/final PDF URL, status, PDF MIME type, 528,115 bytes, 31 pages, and SHA-256. The version of record is Copyright 2015 Department of Mathematics, Princeton University. Current publisher policy materials do not establish the agreement applicable to this 2015 article, so the release conservatively records no article-specific redistribution grant and `redistributable_in_release=false`. The inherited Johnston–Yang author-manuscript lock remains nonredistributable as well.

Declarations

Data and code availability. The proof, certificate compiler, strict tests, closed schema, two remote source-lock records, offline-by-default verification, and release replay tools accompany this paper. The external PDFs and source tar are not redistributed.

Author contributions. The author is responsible for conceptualization, proof, software, validation, writing, and release auditing.

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Ethics. No human participants, animals, personal data, or clinical interventions are involved.

AI assistance. AI-assisted tools were used for exact symbolic-interface enumeration, adversarial testing, manuscript drafting, and release auditing. All mathematical claims, citations, source locks, and final text remain under author responsibility.

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